Chapter 5
A Multidimensional View on Pre-service Teacher Resilience in Germany, Ireland, Malta and Portugal

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Abstract Teacher resilience is a key factor to understand how teachers effectively cope with the strains and stresses of their profession. In this chapter, we are presenting results from a survey study, in four different countries (Germany, Malta, Ireland and Portugal), on pre-service teacher resilience which was part of the project ENTREE (ENhancing Teacher REsilience in Europe). The main goals were to examine the relationships between a global dimension of resilience and its related dimensions and to test the communalities and differences in these relationships amongst the four different countries. Participants were 764 pre-service teachers from Germany, Ireland, Malta and Portugal who answered to a questionnaire including perceptions of individual (e.g. teacher efficacy, commitment, personal life, rumination) and social/contextual (e.g. school support, social context) factors, as well as a global evaluation of resilience. Results showed that self-efficacy appears as one of the key factors related to resilience. Furthermore, differences in the relationships between the variables were found according to each country, suggesting that resilience is influenced by the context in which teachers live and work.

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It is estimated that in many countries nearly half of the early career teachers leave their profession during the first 5 years of teaching (Ingersoll 2003; OECD 2005). The teaching profession has become more and more demanding: teachers experience a high level of pressure to improve standards and performance, a heavy workload, lack of support, inadequate professional preparation and challenging students. These lead to attrition, stress and burnout and are the main reasons for the decision to leave the profession at such an early stage of the career (Cefai and Cavioni 2014; Gu and Li 2013). But what keeps the other half teaching? What helps them not to burnout? What helps them to maintain their "courage to teach" (Gu and Li 2013), to thrive in their career, to ensure quality learning for all students and to contribute to the overall improvement of educational standards? These questions change the perspective in the discussion on teacher retention from those who leave to those who stay (Hong 2012) and move the focus from the more negative concept of teacher burnout to the concept of teacher resilience, where retention is associated with constructs like commitment, engagement and motivation. It focuses on those who continue to develop professionally and deliver high-quality teaching, regardless of their career stage (Day and Gu 2014; Mansfield et al. 2016). Teacher resilience research furthermore tries to find answers on how to help those who struggle by understanding those who not only survive but also thrive.

Teacher resilience emerged as an alternative approach to understand why some teachers remain in their profession, despite the everyday challenges they face. In the literature, it has often been described as a quality for sustaining teacher commitment (Brunetti 2006) or as a dynamic process, where teachers' personal characteristics interact over time with their contextual resources (Mansfield et al. 2012, 2016; Schwarze and Wosnitza 2017, Chap. 3, this volume). This process involves strategies to cope with adverse events (Castro et al. 2010), emotion regulation (Ee and Chang 2010; Tait 2008) and the crucial role that protective and risk factors exert in how individuals respond to adverse events (Cefai and Cavioni 2014; Mansfield et al. 2012).

More recently, resilience has been conceptualised as a multidimensional construct rather than a personal quality (Mansfield et al. 2012). Whilst the concept of resilience from a psychological perspective outlines resilience as a personal quality (Brunetti 2006), the notion of resilience which takes account of the social and cultural contexts of the individual's work (Mansfield et al. 2016) advances a perspective that views resilience as a relative, multidimensional and developmental construct or a dynamic process which encompasses positive adaptation within a given context of significant adversity. In this sense, teacher resilience is not only associated with the capacity to "bounce back" but, rather, is the capacity to maintain equilibrium, a sense of commitment and agency in everyday life and the capacity to manage the unavoidable uncertainties inherent in the realities of teaching (Gu and Day 2013; Schwarze and Wosnitza 2018, Chap. 3, this volume). In this light, Wosnitza et al. (2014) defined teacher resilience as follows:

> Teacher resilience refers to the process of, capacity for, and outcome of positive adaptation as well as on going professional commitment and growth in the face of challenging circumstances. Resilience is shaped by individual, situational and broader contextual characteristics that interrelate in dynamic ways to provide risk (challenging) or protective (supportive)

> factors. Individuals, drawing on personal, professional and social resources, not only "bounce back" but also are able to thrive professionally and personally, experience job satisfaction, positive self-beliefs, personal wellbeing and an on-going commitment to the profession. (p. 2)

In the light of this definition, they propose a heuristic model on teacher resilience, which is based amongst others on Mansfield et al. (2012), who interviewed both graduating and early career teachers and, as a result, proposed a framework that organises the multiple aspects of teacher resilience. A multidimensional model and dynamic perspective has emerged, emphasising the role of professional, motivational, emotional and social aspects of teacher resilience. Aspects of commitment, organisation, flexibility, reflection, time management and teaching skills are central to the professional dimension. The motivational dimension combines features of optimism, positivity and enthusiasm, persistence, confidence, self-efficacy, goal setting, "inner drive" (Hong 2012) and intrinsic motivation (Kitching et al. 2009). Emotions are another key factor of resilience and comprise aspects such as humour (Bobek 2002), emotional management, social and emotional competence (Ee and Chang 2010; Jennings and Greenberg 2009), coping strategies (Castro et al. 2010; Tschannen-Moran and Woolkfolk Hoy 2007) and enjoyment. Finally, the social dimension gathers aspects related to building support and trusting relationships (Cefai and Cavioni 2014), to interpersonal and communication skills, problem-solving and the capacity for seeking help when needed (Tait 2008).

Whilst this framework outlines possible dimensions of teacher resilience, it doesn't take into account the impact of external policy change which has become "a dominant feature in the changing landscape of school governance, curriculum and classrooms" resulting in the fact that "being a teacher and achieving success has become more complex, more highly cognitive and emotionally demanding" (Day and Gu 2014, p. 25). Accordingly, within the context of the project ENTREE[1] (ENhancing Teacher REsilience in Europe), a multilevel heuristic model of teacher resilience was proposed (Wosnitza et al. 2014). In this model, it is postulated that all individual behaviour – "resilient" or "non-resilient" – is a result of an interaction between experiences, skills, knowledge and beliefs and develops in interaction with the environment. This means that every teacher is entering the profession with specific beliefs, capacities, skills and a set of strategies that helps her/him to cope with stressful or challenging situations that may encounter as they adapt to a new context. It is furthermore assumed that teachers can learn how to respond in resilient ways and that resilience can be enhanced over time by specific support systems and environmental factors. Therefore, within this framework, teacher resilience is understood as a multifaceted phenomenon that results from the process of interaction between individual characteristics and contextual factors (see Fig. 5.1).

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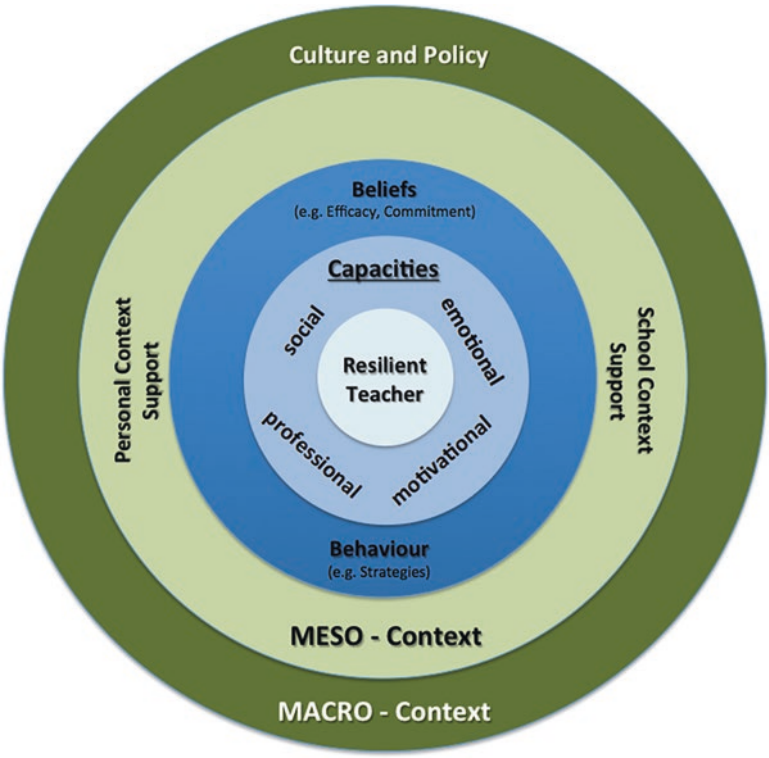

Fig. 5.1 Multilevel model developed by ENTREE (Wosnitza et al. 2014)

At the micro level and as suggested by Mansfield et al. (2012), one finds teachers' motivational, emotional, professional and social capacities. Capacities such as a teacher's intrinsic motivation (Kitching et al. 2009), competence to define professional goals (Sharplin et al. 2011), emotional competence (Chang 2009; Tait 2008) and emotional regulation (Jennings and Greenberg 2009; Morgan 2011; Tugade and Fredrickson 2004), as well as the capacity to adapt to a new social environment and communicate with colleagues and superiors (Schelvis et al. 2014; Sharplin et al. 2011), are considered. On the other hand, understanding resilience means not only identifying existing capacities but also means identifying behaviour dispositions like coping strategies to handle (potentially) stressful and challenging situations in a way that the outcome is satisfying, acceptable or at least manageable. It furthermore means to understand the teacher's beliefs on which her/his behaviour is built.

For this, the model also includes teacher beliefs like self-efficacy or commitment and their behaviour dispositions (e.g. coping strategies). Efficacy beliefs play an important role in teacher resilience (Bobek 2002; Day et al. 2007; Howard and Johnson 2004; Le Cornu 2009) and have also been found to be a strong predictor of teacher resilience (Ee and Chang 2010). The literature acknowledges that teachers with high levels of efficacy also possess high levels of resilience (Tschannen-Moran

and Woolfolk Hoy 2007) and that teachers' efficacy beliefs "have a profound impact on resilience" (Kitching et al. 2009, p. 54). Self-efficacy is regarded as an interactive process and a key component of resilience that can be enhanced as teachers encounter and overcome challenges in their career. Also, teachers' sense of professional commitment is considered as another important factor contributing to their resilience (Day et al. 2007; Gu and Li 2013; Morgan 2011). The literature also underlines the importance of a sense of vocation in helping teachers to promote their personal resources, sustain their motivation and fuel their courage to face and overcome the challenges they meet (Day et al. 2007). Research shows that coping strategies are related to the construct of resilience (Connor and Davidson 2003; Sharplin et al. 2011).

On the meso-level of the model, contextual factors that are crucial for sustaining and maintaining teacher resilience are located. These factors include on the one hand the support a teacher gets at school, like support from colleagues, effective leadership, working in a good school atmosphere and caring and supportive relationships with colleagues (Cefai and Cavioni 2014; Ebersöhn 2012; Howard and Johnson 2004; Le Cornu 2013; Sumsion 2004). On the other hand, they also include the personal support outside school, which a teacher receives from family and friends (Mansfield et al. 2014). Research has shown that a lack of support from school leadership and poor relationships with colleagues, students or parents challenge teachers' capacity of resilience (Gu and Li 2013), outlining the importance of the contextual factors (Cefai 2007; Mansfield et al. 2012) and the personal and professional relationships for positively overcoming challenging situations and sustaining their passion and commitment to teaching (Brunetti 2006; Cefai and Cavioni 2014; Ebersöhn 2012; Morgan 2011; Papatraianou and Le Cornu 2014). The resilience literature also highlights the importance of seeking help or assistance as a coping strategy, when teachers face difficulties in handling the variety of stressors within the school context (Castro et al. 2010; Sharplin et al. 2011; Tait 2008).

Moreover, teachers are confronted with challenges that come with the continuing changes in policy and the increased pressure for teachers to improve their standards, which is a significant factor of stress in their work (Day and Gu 2014; Gu and Li 2013; Price et al. 2012). For this reason, the model proposed by ENTREE (Wosnitza et al. 2014) includes the policy aspects that interfere with teachers' resilience and well-being on the macro level.

Considering teacher resilience as a multidimensional construct and organised in different levels, in this chapter, we sought to identify which factors were most related to teacher resilience. Based on the heuristic model of teacher resilience proposed by ENTREE and presented above, we take into consideration the four levels considered by this model: capacities, beliefs, school and personal supportive contexts and culture and policy. More specifically, we analyse the role of perceived capacities (emotional competence, social competence, motivation, professional flexibility rumination and personal life); teacher commitment and teacher self-efficacy (beliefs); school support, institutional conditions for caring and supportive relationships, engagement in school community and conditions provided by

institution for job satisfaction (school and personal supportive contexts); and administrative and policy demands (culture and policy) in pre-service teachers resilience.

In addition, taking into account the particularities of the teaching reality in different European countries, we aim to analyse differences in the relationship between teachers' resilience and the contributing contextual factors by country level.

Method

Participants

Pre-service teachers (N = 764) from Germany (n = 235, 69.8% female), Ireland (n = 154, 79.1% female), Malta (n = 262, 80.5% female) and Portugal (n = 113, 96.5% female) participated in the study. Participants age ranged from 18 to 48 (M = 25.2, SD = 5.7).

Measures

Measures used in this study comprised a global measure of teacher resilience, the Multidimensional Teacher Resilience Scale and scales measuring personal life, rumination, teacher commitment, teacher self-efficacy, social context, school support, conditions provided by institutions for job satisfaction and administrative and policy demands. In the following section, we briefly present each measure.

Global Measure of Teacher Resilience To assess resilience, we used the Teacher Resilience Scale (Morgan 2011), which encompasses nine items intended to assess how teachers deal with setbacks in school (e.g. "While things can upset me, I find I can bounce back"). Instead of the original 7-point scale, in this study, participants were asked to evaluate their level of confidence in a 5-point scale ranging from "Not confident at all" to "Absolutely confident".

Multidimensional Teacher Resilience Scale The Multidimensional Teacher Resilience Scale (MTRS) (Mansfield and Wosnitza 2015) comprises 16 items intended to assess 4 dimensions of teacher resilience: emotional (5 items, e.g. "When I feel upset or angry at school I can manage to stay calm"), social (4 items, e.g. "I am good at building relationships in new school environments"), motivational (8 items, e.g. "I like challenges in my work") and professional (2 items, e.g. "I can quickly adapt to new situations at school"). All items were answered in a 5-point Likert scale ranging from "completely agree" to "completely disagree".

Personal Life This is a four-item measure adapted from Morgan (2011) intended to assess the satisfaction with life outside school (e.g. "I have been happy with my

social and personal life"). Participants' responses to each item were scored on a 5-point Likert scale ("strongly disagree" to "strongly agree").

Rumination The rumination scale (Morgan 2011) comprises four items assessing the extent to which teachers continue to think about problems that occurred at school (e.g. "When something upsets me at school, I find it hard to forget about it"). The response scale is a 5-point frequency scale ("Never" to "All the time").

Teacher Commitment Commitment is an eight-item measure adapted from Morgan (2011), assessing both commitment to profession (vocational commitment, e.g. "I feel that teaching is really right for me") and commitment to the school where teacher works (organisational commitment, e.g. "I am happy to continue working in this school"). Answers were given in a 5-point Likert scale ranging from "completely agree" to "completely disagree".

Teacher Self-Efficacy The Teacher Self-Efficacy (TSE) Scale was based on Morgan (2011), to which we added some new items assessing the feelings of competence in teaching (four items, e.g. "How confident you feel: Explaining difficult material in ways that the children will understand it") and behaviour management (four items, e.g. "How confident you feel: Managing inappropriate behaviour"). Items were rated on a 5-point scale ranging from "Not confident at all" to "Absolutely confident".

Social Context Social context is a three-dimensional measure adapted from Cefai and Cavioni (2014) intended to assess the perception of conditions provided by the school in terms of caring and supportive relationships (ICCSR) (nine items, e.g. "Our staff understands, supports and cares for one other"), the perception of support provided by the school administration (SPSA) (three items, e.g. "I am understood and supported by the administration" and the perception of collaborative partnerships in the school community (six items, e.g. "There are a number of teaching partnerships and/or mentoring schemes in our school"). Responses ranged on a 5-point Likert scale from "completely agree" to "completely disagree".

School Support The school support measure (Morgan 2011) has four items addressing the perception of support from colleagues at school (e.g. "When something goes wrong I can talk to some of the other teachers"). Items are rated on a 5-point Likert scale ranging from "Never" to "All the time".

Conditions Provided by Institutions for Job Satisfaction (CPIJS) This is a 12-item measure (Silva 2013) aiming to assess the perception of the presence of job satisfaction conditions in three dimensions: resources and working conditions (e.g. "To what extent these factors of job satisfaction are present in your teaching practice? School working conditions"), recognition and professional development (e.g. "To what extent these factors of job satisfaction are present in your teaching practice? Recognition and appreciation of teachers' performance by the school community")

and school organisational climate (e.g. "To what extent these factors of job satisfaction are present in your teaching practice? Support and monitoring provided by school administration"). Participants rate each item on a 5-point scale ranging from "Never" to "Always". In this study, we used the global measure of job satisfaction by averaging the 12 items.

Administrative and Policy Demands This measure comprises six items assessing the perception of stress induced by changes in educational policies and in the educational system (e.g. "I find it frustrating that as soon as I become familiar with one policy directive another one replaces it"). The items were answered in a 5-point scale ranging from "Always" to "Never".

Procedure

All measures were included in the survey, which was hosted online between April and July 2014. In Portugal and Ireland, participants were recruited through email directed to higher education institutions that award teaching degrees. As an incentive, participants were entered into a prize draw to win a voucher for an online bookstore. In Germany, participants were recruited using two approaches: some teacher training students at a university were contacted via email and asked to do the online survey; teacher training students and pre-service teachers in the practical phase of their training were also asked to fill in the paper and pencil survey. In Malta, the online questionnaire was sent to all students following the Bachelor of Education course at a Faculty of Education.

At the beginning of the survey, all participants were informed about the aims of the project. Confidentiality and anonymity were assured since no identifying data were collected.

Data Analysis

For all the measures, confirmatory factor analyses (CFA) were carried out to confirm the factor structures. All CFA were conducted using the maximum likelihood estimator available in AMOS 24.0 (Arbuckle 2016). The models' goodness of fit was assessed using CFI, TLI and RMSEA. For CFI and TLI values, above 0.90 were considered as indicating a good fit of the models to the data, whereas for RMSEA values below 0.08 indicate an acceptable fit (Kline 2011). To assess reliability we used Cronbach's alpha.

Following the model proposed by the ENTREE project, we carried out a hierarchical regression analysis considering the global measure of resilience as dependent variable and all the other measures as independent variables. In the first step, we included the variables related to capacities: the four dimensions of the Multidimensional Teacher Resilience Scale (emotional competence, social compe-

tence, motivation and professional flexibility), personal life and rumination. In the second block, we added the variables related to the second level of the ENTREE model: the domain of beliefs. Thus, we included in the analysis the two dimensions of teacher commitment (profession and school) and the two dimensions of self-efficacy: self-efficacy for teaching and for behaviour management. In the third step, we added the variables considered in the meso-context, namely, those related to the school and personal supportive contexts (e.g. the two dimensions of social context measure – institutional conditions for caring and supportive relationships and engagement in school community; school support and the perception of conditions provided by institution for job satisfaction). The last block comprised the perception of administrative and policy demands as well as the variables considered previously. All measures were computed averaging the items.

Results

Confirmatory factor analyses were carried out for each measure separately showing an adequate fit to the data (CFI ranging from 0.935 to 0.999; TLI ranging from 0.922 to 0.996, respectively, for the Multidimensional Teacher Resilience Scale and the Personal Life Scale; and RMSEA ranging from 0.032 to 0.102, for the Personal Life scale and the Teacher Commitment Scale).

Table 5.1 shows the descriptive statistics, reliabilities for all the variables in the study and the correlations between them. With the exception of policy demands and rumination, all the variables relate positively with the global measure of resilience. Correlations with the global measure of resilience ranged from small to medium with the two dimensions of self-efficacy showing the highest correlations with the global measure of resilience.

Table 5.2 shows the results of the hierarchical regression analysis on the global measure of resilience. The first block of variables explains 27.4% of variance of teachers' resilience. From the variables included in this first step of analysis, social competence, professional flexibility and personal life are significantly associated with teachers' resilience. Concerning the variables added in the second step (teacher commitment and self-efficacy), only the dimensions of self-efficacy were significantly related to teachers' resilience. Moreover, when adding the beliefs-related variables, a significant association between rumination and resilience was found. This second block increased the explained variance of teachers' resilience by 13%. The third block has taken into consideration the variables related to schools' and personal supportive contexts. Adding those variables did not add much to the variance explained in teachers' resilience ($\Delta R^2 = 0.032$). However, the addition of the variables in the third block suppressed the significant association between professional flexibility and resilience but increased the explanation of emotional competence and teacher commitment in global resilience. Engagement in school community, school support and conditions provided by institution for job satisfaction are the variables added in the third step that were significantly related to resilience.

Table 5.1 Descriptive statistics, reliabilities and Pearson correlations between the variables of the study

	Variables	M	SD	Observed range	Cronbach's alpha	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	Resilience (global)	2.18	0.59	1–5	0.89																
2	MTRS – emotional competence	2.17	0.65	1–4.5	0.68	0.31															
3	MTRS – social competence	1.91	0.54	1–4.25	0.67	0.39	0.32														
4	MTRS – motivation	1.93	0.53	1–5	0.78	0.29	0.60	0.39													
5	MTRS – professional flexibility	2.01	0.70	1–5	0.75	0.36	0.38	0.36	0.37												
6	Personal life	1.90	0.79	1–4.75	0.83	0.26	0.29	0.18	0.27	0.12											
7	Rumination	2.22	0.86	1–5	0.87	−0.09	−0.18	−0.02	−0.06	−0.07	−0.05										
8	Teacher commitment – profession	1.77	0.74	1–5	0.88	0.32	0.32	0.26	0.39	0.28	0.34	−0.03									

9	Teacher commitment – school	1.98	0.93	1–5	0.89	0.27	0.28	0.23	0.28	0.28	0.24	−0.07	0.52								
10	TSE – teaching	2.08	0.63	1–4.25	0.82	0.45	0.20	0.28	0.28	0.28	0.17	0.07	0.22	0.13							
11	TSE – behaviour management	2.09	0.68	1–4.75	0.81	0.45	0.22	0.30	0.25	0.29	0.15	0.06	0.23	0.17	0.59						
12	ICCSR	2.22	0.70	1–4.67	0.90	0.32	0.26	0.36	0.28	0.31	0.24	−0.06	0.24	0.52	0.18	0.19					
13	SPSA	2.30	0.83	1–5	0.85	0.22	0.17	0.19	0.25	0.15	0.09	0.04	0.08	0.33	0.13	0.19	0.43				
14	Engagement in school community	2.43	0.67	1–4.67	0.77	0.40	0.30	0.39	0.35	0.41	0.17	−0.05	0.31	0.36	0.25	0.33	0.48	0.34			
15	School support	2.06	0.73	1–5	0.86	0.34	0.17	0.22	0.17	0.30	0.23	−0.04	0.21	0.42	0.19	0.16	0.64	0.27	0.41		
16	CPIJS	2.51	0.69	1–5	0.91	0.33	0.07	0.25	0.12	0.26	0.13	0.03	0.21	0.24	0.27	0.36	0.28	0.13	0.35	0.30	
17	Policy demands	2.63	0.71	1–5	0.81	−0.11	−0.07	−0.06	−0.06	−0.06	−0.06	0.22	−0.06	−0.04	−0.04	−0.02	−0.01	−0.06	−0.04	−0.09	0.06

MTRS multidimensional teacher resilience scale, *TSE* teacher self-efficacy, *ICCSR* institutional conditions for caring and supportive relationships, *SPSA* support provided by the school administration, *CPIJS* conditions provided by institution for job satisfaction. Correlations higher than 0.06 are significant at $p < 0.05$. Correlations higher than 0.09 are significant at $p < 0.01$

Table 5.2 Hierarchical multiple regression analyses for predictors of teachers' resilience scores

		Block 1			Block 2			Block 3			Block 4		
	Variables	*B*	*SE*	β	*B*	*SE*	β	*B*	*SE*	β	*B*	*SE*	β
Competencies	MTRS – emotional competence	0.08	0.04	0.09	0.06	0.04	0.07	0.08*	0.04	0.08*	0.08*	0.04	0.09*
	MTRS – social competence	0.29**	0.04	0.26**	0.20**	0.04	0.18**	0.16**	0.04	0.15**	0.16**	0.04	0.15**
	MTRS – motivation	0.02	0.05	0.02	−0.06	0.05	−0.06	−0.07	0.05	−0.07	−0.08	0.05	−0.07
	MTRS – professional flexibility	0.17**	0.03	0.20**	0.09**	0.03	0.11**	0.05	0.03	0.06	0.05	0.03	0.06
	Personal life	0.12**	0.03	0.15**	0.07**	0.03	0.09**	0.05*	0.03	0.07*	0.05*	0.03	0.07*
	Rumination	−0.04	0.03	−0.05	−0.06**	0.02	−0.09**	−0.06**	0.02	−0.09**	−0.06*	0.02	−0.08*
Beliefs	Teacher commitment – profession				0.06	0.03	0.08	0.07*	0.03	0.09*	0.07*	0.03	0.09*
	Teacher commitment – school				0.05	0.03	0.07	−0.01	0.03	−0.02	−0.01	0.03	−0.01
	Teacher self-efficacy – teaching				0.21**	0.04	0.22**	0.20**	0.04	0.21**	0.20**	0.04	0.21**
	Teacher self-efficacy – behaviour management				0.17**	0.04	0.19**	0.14**	0.04	0.16**	0.14**	0.04	0.16**
Meso-context	Institutional conditions for caring and supportive relationships							−0.02	0.04	−0.03	−0.02	0.04	−0.02
	Support provided by the school administration							0.04	0.03	0.05	0.03	0.04	0.05
	Engagement in school community							0.08*	0.04	0.09*	0.08*	0.04	0.09*
	School support							0.10**	0.04	0.13**	0.10**	0.04	0.12**
	Conditions provided by institution for job satisfaction							0.07*	0.03	0.08*	0.07*	0.03	0.08*
Macro-context	Policy demands										−0.03	0.03	−0.04
	R^2	0.249			0.379			0.411			0.413		
	F	33.4			36.5			27.6			26.0		
	ΔR^2	0.249			0.13			0.032			0.002		

MTRS multidimensional teacher resilience scale
$^*p < 0.05$; $^{**}p < 0.01$

In the last step, we added the perception of administrative and policy demands, which was not a significant predictor of pre-service teachers' resilience (Table 5.2).

In the final model, the strongest predictors of teachers' resilience were the two dimensions of self-efficacy (teaching and behaviour management) and social competence. Perception of school support, teacher commitment, emotional competence and engagement in school community are the variables that follow in the magnitude of the association with teachers' resilience. Finally, the only predictor that relates negatively and significantly with resilience is rumination, meaning that continuing to think about problems after their occurrence related negatively with resilience.

Four separate hierarchical regression analyses, one for each country, were carried out using only the variables of the two first levels of the model (competencies and beliefs) since data collection in Portugal did not include the measures for the meso- and macro-contexts. Table 5.3 shows the results for the last model tested in the hierarchical regression analyses for each country. For German pre-service teachers, the strongest predictors of resilience are teacher self-efficacy – teaching, emotional competence and teacher self-efficacy – behaviour management. The strongest predictor of Portuguese pre-service teachers' resilience is motivation followed by emotional competence and the teaching dimension of self-efficacy. For Maltese pre-service teachers, the strongest predictor of resilience is personal life followed by the teaching dimension of self-efficacy, social and emotional competence and self-efficacy in behaviour management. The two dimensions of self-efficacy are the only variables significantly related to resilience in Irish pre-service teachers.

Differences also arise when we look at the variance explained in resilience, with 55% of variance explained in the Portuguese sample, which is a value substantively higher than in the other samples.

Discussion

The first aim of this study was to analyse the factors that explain pre-service teachers' resilience. Based on the theoretical model developed by the ENTREE team (Wosnitza et al. 2013), we tested it through hierarchical regression analysis. Findings support the hypothesised model showing significant associations between variables at three of the four levels of the model and resilience. The results confirm the conception of resilience as a multidimensional construct, involving a dynamic relation between personal and contextual factors (Cefai and Cavioni 2014; Mansfield et al. 2012, 2016). On the one hand, it highlighted the relationship between social and professional competence and the capacity of resilience (Mansfield et al. 2012) but also the importance of personal life to deal with the demands of teacher profession. Also, being able to reflect about teacher experience and practice (Le Cornu 2009) and being highly committed to students are shown to have a significant impact in teacher resilience. When adding new personal variables in the model, the results show a significant impact of commitment and self-efficacy (both in teaching and behaviour management) in teacher resilience.

Table 5.3 Hierarchical multiple regression analyses for predictors of teachers' resilience scores

	German			Portuguese			Maltese			Irish		
Variables	*B*	*SE*	β	*B*	*SE*	β	*B*	*SE*	β	*B*	*SE*	β
MTRS – emotional competence	0.18**	0.06	0.21**	0.28*	0.11	0.26*	0.14*	0.06	0.14*	−0.03	0.09	−0.04
MTRS – social competence	0.10	0.07	0.09	0.16	0.11	0.14	0.15*	0.06	0.15*	0.09	0.10	0.09
MTRS – motivation	0.08	0.08	0.07	0.35*	0.14	0.27*	−0.04	0.07	−0.04	0.16	0.13	0.14
MTRS – professional flexibility	0.04	0.05	0.05	0.06	0.09	0.07	0.05	0.05	0.06	0.02	0.08	0.02
Personal life	0.05	0.04	0.10	[a]	[a]	[a]	0.16**	0.05	0.19**	0.07	0.07	0.08
Rumination	−0.02	0.04	−0.04	−0.09	0.07	−0.11	−0.07	0.04	−0.09	−0.03	0.04	−0.05
Teacher commitment – profession	0.03	0.05	0.04	0.14	0.09	0.17	0.07	0.05	0.09	0.05	0.08	0.05
Teacher commitment – school	0.04	0.03	0.09	0.13	0.07	0.19	0.05	0.04	0.07	0.07	0.08	0.07
Teacher self-efficacy – teaching	0.16*	0.06	0.18*	0.19*	0.09	0.20*	0.16**	0.06	0.18**	0.28**	0.09	0.32**
Teacher self-efficacy – behaviour management	0.18**	0.06	0.23**	0.06	0.09	0.07	0.11*	0.05	0.14*	0.20*	0.09	0.22*
R^2	0.39			0.55			0.37			0.44		
F	11.7			12.6			14.8			10.8		

MTRS multidimensional teacher resilience scale
*$p < 0.05$; **$p < 0.01$
[a]For Portuguese sample, data collection did not include the Personal Life measure

A sense of commitment, purpose, agency and vocation have been considered as aspects contributing to teacher effectiveness and fostering resilience (Day et al. 2007; Gu and Li 2013; Hong 2012; Morgan 2011; Tait 2008). Self-efficacy, globally, appears as the main predictor of teachers' sense of resilience. In previous studies (e.g. Bobek 2002; Ee and Chang 2010; Hong 2012; Howard and Johnson 2004), it was found that the beliefs about efficacy were reported to be the major components in teacher resilience and that self-beliefs about classroom management often constitute a personal risk factor (Beltman et al. 2011; Gu and Day 2013; Kitching et al. 2009). School support and engagement in school community were the contextual factors related to teacher resilience. In fact, when adding these aspects, a decline occurred in the significance of the previous factors, revealing that these contextual variables relate to personal resources (cf. Cefai and Cavioni 2014). Also, it gives strength to the conception of resilience as involving the context where teachers teach, rather than being an innate trait (Brunetti 2006). These results reveal the vital role of school organisation environment and support by colleagues and school leaders (Carmel and Cavioni 2014; Ebersöhn 2012; Gu and Day 2013; Le Cornu 2013; Papatraianou and Le Cornu 2014) in fostering teacher resilience and well-being and successful learning outcomes for all students (Gu and Day 2013).

Finally, no significant relationship was found between administrative and policy demands and teacher resilience, which can be explained by the fact that participants were pre-service teachers, who had not yet experienced the policy demands and the increased accountability imposed on teachers.

The second aim of the study was to analyse differences in the relationships between the variables considered and resilience in the various countries. The comparison of the pattern of associations between the variables considered and resilience showed similarities between the countries as well as some differences. Thus, despite self-efficacy being an influential predictor across the four countries as the global analysis showed, only for the Irish and German teachers it is the strongest predictor, whereas for the remaining countries it is emotional competence (Portuguese) or personal life (Maltese) that has the strongest relation with resilience.

Based on the evidence of these results, teacher resilience is not an innate state, but, rather, it is constituted and developed by the dynamic interaction of multiple aspects of teachers' personal and contextual competencies and resources that can be organised in different levels, from micro-level factors such as personal competencies to macro-level factors, acknowledging the crucial impact that the increased demands and policies exert on the teaching profession.

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